```
['.config', 'sample_data']
```

```
['.config', 'sample_data']
```

```
import pandas as pd
df = pd.read_csv("/content/penguins.csv")
df.head()
```

	species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	body
0	Adelie	Torgersen	39.1	18.7	181.0	
1	Adelie	Torgersen	39.5	17.4	186.0	
2	Adelie	Torgersen	40.3	18.0	195.0	
3	Adelie	Torgersen	NaN	NaN	NaN	
4	Adelie	Torgersen	36.7	19.3	193.0	

Next steps: Generate code with df | New interactive sheet

```
import os

print(os.listdir("/content"))
```

```
['.config', 'penguins.csv', 'sample_data']
```

```
import pandas as pd

df = pd.read_csv("penguins.csv")

df.head()
```

	species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	body
0	Adelie	Torgersen	39.1	18.7	181.0	
1	Adelie	Torgersen	39.5	17.4	186.0	
2	Adelie	Torgersen	40.3	18.0	195.0	
3	Adelie	Torgersen	NaN	NaN	NaN	
4	Adelie	Torgersen	36.7	19.3	193.0	

Next steps: Generate code with df | New interactive sheet

```
df.shape
```

```
(344, 8)
```

```
print(df.shape)
```

```
(344, 8)
```

```
df.isna().sum()
```

	0
species	0
island	0
bill_length_mm	2
bill_depth_mm	2
flipper_length_mm	2
body_mass_g	2
sex	11
year	0

dtype: int64

```
df["species"].value_counts()
```

species	count
Adelie	152
Gentoo	124
Chinstrap	68

dtype: int64

```
df.dtypes
```

	0
species	object
island	object
bill_length_mm	float64
bill_depth_mm	float64
flipper_length_mm	float64
body_mass_g	float64
sex	object
year	int64

dtype: object

Gemini

```
# 1. Import necessary libraries for visualization
import matplotlib.pyplot as plt
import seaborn as sns

# 2. Print the missing-value count for each column
# We use the .isnull().sum() method to find out how many missing (NaN) val
print("--- Missing Values Per Column ---")
missing_data = df.isnull().sum()
print(missing_data)
print()

# 3. Calculate and print mean flipper_length_mm by species
# pandas.groupby automatically excludes missing (NaN) values from the calc
# ensuring we get an accurate average for each species.
print("--- Mean Flipper Length by Species ---")
mean_flipper = df.groupby("species")["flipper_length_mm"].mean()
print(mean_flipper)
print()
print("Verification Hint: You can verify this by filtering the DataFrame f
print("individually (e.g., df[df['species'] == 'Adelie']['flipper_length_m

# 4. Create a labeled seaborn boxplot of flipper_length_mm by species
# We set the overall background style first
sns.set_theme(style="whitegrid")

# Create a figure with custom dimensions (width=8 inches, height=6 inches)
plt.figure(figsize=(8, 6))

# Create the boxplot. We map 'hue' to 'species' and set 'legend=False' to
sns.boxplot(data=df, x="species", y="flipper_length_mm", hue="species", pa
# Create the boxplot with our custom mean markers configured as green tria
# We set x and hue to 'species', and legend to False to avoid warnings.
sns.boxplot(
    data=df,
    x="species",
    y="flipper_length_mm",
    hue="species",
```

```
    palette="Set2",
    legend=False,
    showmeans=True,
    meanprops={
        "marker": "^",
        "markerfacecolor": "forestgreen",
        "markeredgecolor": "black",
        "markersize": "8"
    }
)

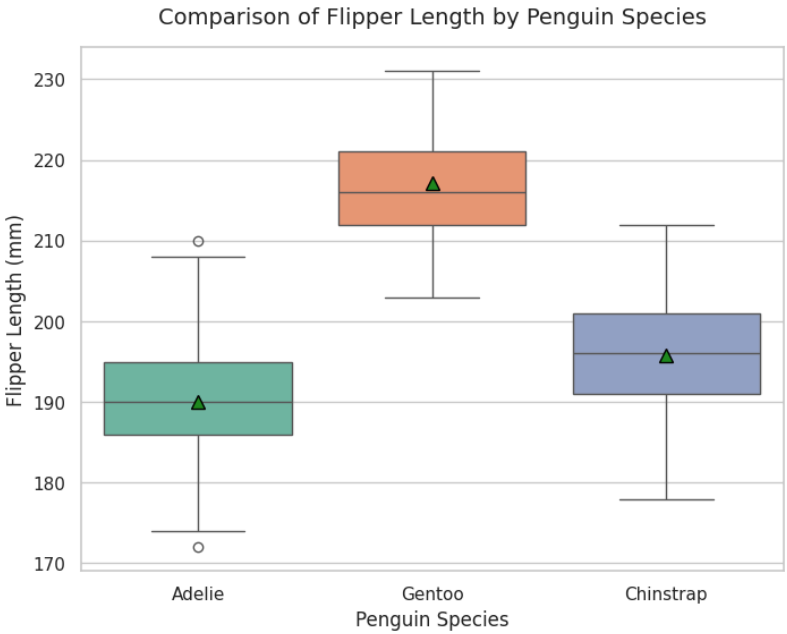
# 5. Add beginner-friendly labels and a title
plt.title("Comparison of Flipper Length by Penguin Species", fontsize=14,
plt.xlabel("Penguin Species", fontsize=12)
plt.ylabel("Flipper Length (mm)", fontsize=12)

# Render the complete chart on the screen
plt.show()
```

```
--- Missing Values Per Column ---
species               0
island                0
bill_length_mm        2
bill_depth_mm         2
flipper_length_mm     2
body_mass_g           2
sex                  11
year                  0
dtype: int64

--- Mean Flipper Length by Species ---
species
Adelie       189.953642
Chinstrap    195.823529
Gentoo       217.186992
Name: flipper_length_mm, dtype: float64

Verification Hint: You can verify this by filtering the DataFrame for each
individually (e.g., df[df['species'] == 'Adelie']['flipper_length_mm'].mean
```




```
df[df["species"] == "Adelie"]["flipper_length_mm"].mean()
```

```
np.float64(189.95364238410596)
```

```
# Average flipper length for Chinstrap penguins
df[df["species"] == "Chinstrap"]["flipper_length_mm"].mean()
```

```
np.float64(195.8235294117647)
```

```
# Average flipper length for Gentoo penguins
df[df["species"] == "Gentoo"]["flipper_length_mm"].mean()
```

```
np.float64(217.1869918699187)
```

```
Start coding or generate with AI.
```

Step 5: Verify AI Results

I independently verified the mean flipper length for each penguin species using pandas filtering and the `.mean()` method.

- **Adelie:** 189.95 mm
- **Chinstrap:** 195.82 mm
- **Gentoo:** 217.19 mm

My independent calculations matched the results from the AI-generated group-by analysis. This increases my confidence that the code calculates the averages correctly. Pandas automatically ignores missing flipper-length values when calculating the mean.

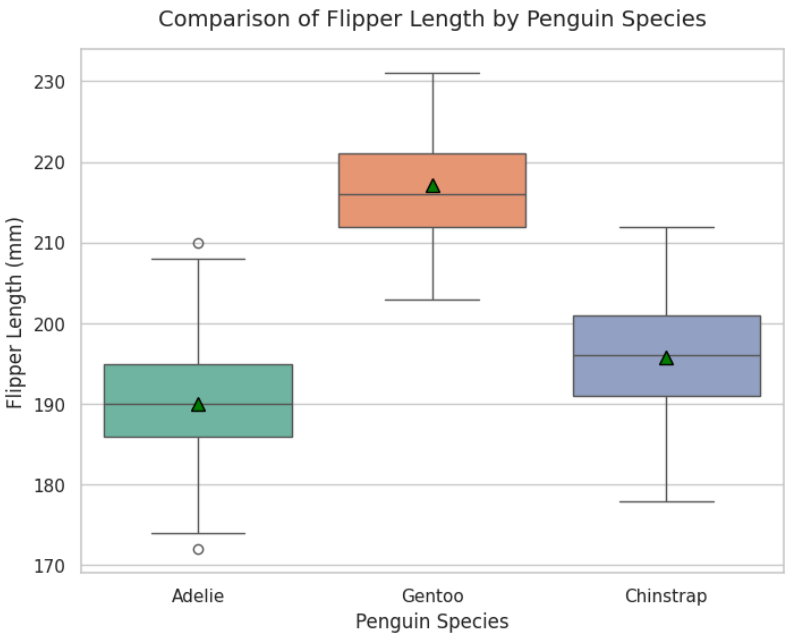
Double-click (or enter) to edit

```
# Create a boxplot with mean markers
plt.figure(figsize=(8, 6))

sns.boxplot(
    data=df,
    x="species",
    y="flipper_length_mm",
    hue="species",
    palette="Set2",
    legend=False,
    showmeans=True,
    meanprops={
        "marker": "^",
        "markerfacecolor": "green",
        "markeredgecolor": "black",
        "markersize": 9
    }
)

# Add the title and labels
plt.title("Comparison of Flipper Length by Penguin Species", fontsize=14,
plt.xlabel("Penguin Species", fontsize=12)
plt.ylabel("Flipper Length (mm)", fontsize=12)
```

```
# Display the chart
plt.show()
```




```
Start coding or generate with AI.
```

Step 6: Evaluate an AI-Suggested Improvement

I asked AI to improve my boxplot by adding a green triangle to represent the mean flipper length for each penguin species. I accepted this suggestion and ran the updated code successfully.

The green triangles make it easier to compare the mean with the median line inside each box. This improves the chart because readers can see both measures of central tendency. I kept the improvement because it makes the visualization more informative.

Double-click (or enter) to edit

```
217.1869918699187 - 189.95364238410596
```

```
27.233349485812738
```

```
Start coding or generate with AI.
```

Double-click (or enter) to edit